

# Supplementary Appendix

## Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

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# 1 Model Structure and Assumptions

## 1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

| State | Clinical Definition                                                                                                                     | Utility | Monthly Cost |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|---------|--------------|
| RF    | Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses) | 0.83    | \$0          |
| LR    | Confirmed locoregional recurrence without distant metastasis; includes salvage surgery, re-treatment, and surveillance                  | 0.64    | \$3,000      |
| DM    | Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring                         | 0.55    | \$12,000     |
| Death | All-cause mortality                                                                                                                     | 0       | \$0          |

*Note:* Utilities and costs are drawn from published melanoma CEA literature [? ? ?]. Costs are in 2026 USD. See Supplementary Table S12 for full PSA distribution parameters.

## 1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time  $t$ .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time  $t$ .
- **OS(t)** — Overall survival: probability of being alive at time  $t$ .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive:  $RF = RFS$ .
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free:  $LR = DMFS - RFS$ .
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free:  $DM = OS - DMFS$ .
- The dead state is the complement of overall survival:  $Dead = 1 - OS$ .

The constraint  $RFS \leq DMFS \leq OS$  is enforced at all time points to ensure non-negative state occupancy. This ordering is clinically meaningful: a patient cannot have distant metastasis without having had a recurrence event ( $RFS \leq DMFS$ ), and a patient cannot die without having both events ( $DMFS \leq OS$ ). The ratio-constrained structure  $r_{LR} = RFS/DMFS$  and  $r_{DM} = DMFS/OS$  guarantees this ordering by construction.

### 1.3 Model Schematic

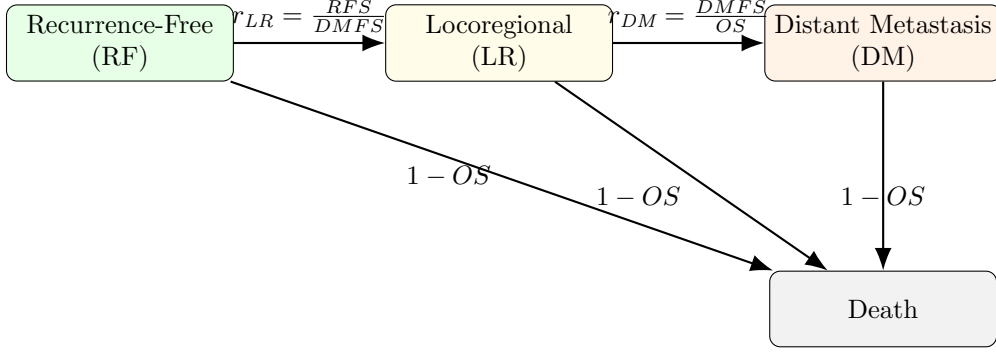


Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship  $RFS \leq DMFS \leq OS$ .

## 2 Survival Data and Parametric Extrapolation

### 2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [?] using WebPlotDigitizer v4.7. Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS. The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

| Month | RFS (%) |        | DMFS (%) |        | OS (%) |        |
|-------|---------|--------|----------|--------|--------|--------|
|       | Combo   | Pembro | Combo    | Pembro | Combo  | Pembro |
| 18    | 80.4    | 62.2   | 92.0     | 73.0   | 96.0   | 93.4   |
| 24    | 78.3    | 60.0   | 90.8     | 70.6   | 96.0   | 93.4   |
| 36    | 73.8    | 55.6   | 85.6     | 65.4   | 93.8   | 85.6   |
| 48    | 72.4    | 49.1   | 83.9     | 65.4   | 92.2   | 85.6   |
| 60    | 68.8    | 49.1   | —        | —      | 92.2   | 71.3   |

**Digitization method:** WebPlotDigitizer v4.7, manual extraction. The 60-month landmark values match the published 5-year ASCO 2026 update [?]. At earlier time points, the 18-month RFS rates from the primary analysis [?] differ from digitized values by 1.4 pp (combo) and 0.2 pp (pembro), consistent with typical digitization error. The 5-year RFS 95% CIs: combo 56.3–78.3%, pembro 33.3–63.0%.

### 3 Clinical Input Provenance

Table S3 summarizes the provenance of all clinical inputs used in the model. The evidence status column classifies each input into one of five categories:

- **Published:** The value is directly reported in the cited source.
- **Digitized:** The value was extracted from a published Kaplan–Meier curve using Web-PlotDigitizer v4.7.
- **Derived:** The value is mathematically derived from other inputs (e.g., ratio of survival functions).
- **Modeled:** The value is produced by the model’s parametric extrapolation (e.g., long-term survival beyond observed data).
- **Topline only:** The source reports a statistically significant result but the numerical hazard ratio or landmark survival probabilities have not yet been disclosed.

Table S3: Clinical Input Provenance

| Input                                           | Value      | Evidence Status | Source               |
|-------------------------------------------------|------------|-----------------|----------------------|
| Input                                           | Value      | Evidence Status | Source               |
| <i>KEYNOTE-942 (Phase 2b, 5-year follow-up)</i> |            |                 |                      |
| <b>Recurrence-Free Survival (RFS)</b>           |            |                 |                      |
| 18-month RFS, combo                             | 80.4%      | Digitized       | Fig. 1, Khattak 2026 |
| 24-month RFS, combo                             | 78.3%      | Digitized       | Fig. 1, Khattak 2026 |
| 36-month RFS, combo                             | 73.8%      | Digitized       | Fig. 1, Khattak 2026 |
| 48-month RFS, combo                             | 72.4%      | Digitized       | Fig. 1, Khattak 2026 |
| 60-month RFS, combo                             | 68.8%      | Published       | Khattak 2026         |
| 60-month RFS, combo 95% CI                      | 56.3–78.3% | Published       | Khattak 2026         |
| 18-month RFS, pembro                            | 62.2%      | Digitized       | Fig. 1, Khattak 2026 |
| 24-month RFS, pembro                            | 60.0%      | Digitized       | Fig. 1, Khattak 2026 |
| 36-month RFS, pembro                            | 55.6%      | Digitized       | Fig. 1, Khattak 2026 |
| 48-month RFS, pembro                            | 49.1%      | Digitized       | Fig. 1, Khattak 2026 |
| 60-month RFS, pembro                            | 49.1%      | Published       | Khattak 2026         |
| 60-month RFS, pembro 95% CI                     | 33.3–63.0% | Published       | Khattak 2026         |
| <b>Distant Metastasis-Free Survival (DMFS)</b>  |            |                 |                      |
| 18-month DMFS, combo                            | 92.0%      | Digitized       | Fig. 1, Khattak 2026 |
| 24-month DMFS, combo                            | 90.8%      | Digitized       | Fig. 1, Khattak 2026 |
| 36-month DMFS, combo                            | 85.6%      | Digitized       | Fig. 1, Khattak 2026 |
| 48-month DMFS, combo                            | 83.9%      | Digitized       | Fig. 1, Khattak 2026 |
| 60-month DMFS, combo                            | –          | Not reported    | N/A                  |
| 18-month DMFS, pembro                           | 73.0%      | Digitized       | Fig. 1, Khattak 2026 |
| 24-month DMFS, pembro                           | 70.6%      | Digitized       | Fig. 1, Khattak 2026 |
| 36-month DMFS, pembro                           | 65.4%      | Digitized       | Fig. 1, Khattak 2026 |
| 48-month DMFS, pembro                           | 65.4%      | Digitized       | Fig. 1, Khattak 2026 |
| 60-month DMFS, pembro                           | –          | Not reported    | N/A                  |
| <b>Overall Survival (OS)</b>                    |            |                 |                      |

| Input                                                | Value                     | Evidence Status | Source                   |
|------------------------------------------------------|---------------------------|-----------------|--------------------------|
| 18-month OS, combo                                   | 96.0%                     | Digitized       | Fig. 1, Khattak 2026     |
| 24-month OS, combo                                   | 96.0%                     | Digitized       | Fig. 1, Khattak 2026     |
| 36-month OS, combo                                   | 93.8%                     | Digitized       | Fig. 1, Khattak 2026     |
| 48-month OS, combo                                   | 92.2%                     | Published       | Khattak 2026             |
| 60-month OS, combo                                   | 92.2%                     | Published       | Khattak 2026             |
| 18-month OS, pembro                                  | 93.4%                     | Digitized       | Fig. 1, Khattak 2026     |
| 24-month OS, pembro                                  | 93.4%                     | Digitized       | Fig. 1, Khattak 2026     |
| 36-month OS, pembro                                  | 85.6%                     | Digitized       | Fig. 1, Khattak 2026     |
| 48-month OS, pembro                                  | 85.6%                     | Published       | Khattak 2026             |
| 60-month OS, pembro                                  | 71.3%                     | Published       | Khattak 2026             |
| <b>Hazard Ratios</b>                                 |                           |                 |                          |
| RFS HR (combo vs. pembro)                            | 0.510                     | Published       | Khattak 2026             |
| RFS HR 95% CI                                        | 0.294–0.887               | Published       | Khattak 2026             |
| DMFS HR (combo vs. pembro)                           | 0.411                     | Published       | Khattak 2026             |
| DMFS HR 95% CI                                       | 0.200–0.843               | Published       | Khattak 2026             |
| OS HR (combo vs. pembro)                             | 0.471                     | Published       | Khattak 2026             |
| OS HR 95% CI                                         | 0.165–1.345               | Published       | Khattak 2026             |
| OS events                                            | 14 (7 per arm)            | Published       | Khattak 2026             |
| <i>INTerpath-001 (Phase 3, topline announcement)</i> |                           |                 |                          |
| RFS                                                  | Statistically significant | Topline only    | Merck/Moderna 2026       |
| DMFS                                                 | Statistically significant | Topline only    | Merck/Moderna 2026       |
| Numerical HR / KM curves                             | Not yet disclosed         | Topline only    | N/A                      |
| <i>Modeled / Derived Values</i>                      |                           |                 |                          |
| <b>Fitted log-normal parameters</b>                  |                           |                 |                          |
| OS $\mu$ (combo), $\sigma$ (combo)                   | 8.396, 3.096              | Derived         | Least-squares fit        |
| OS $\mu$ (pembro), $\sigma$ (pembro)                 | 4.841, 1.131              | Derived         | Least-squares fit        |
| r_DM $\mu$ (combo), $\sigma$ (combo)                 | 7.113, 2.480              | Derived         | Ratio fit                |
| r_DM $\mu$ (pembro), $\sigma$ (pembro)               | 7.042, 5.000              | Derived         | Ratio fit                |
| r_LR $\mu$ (combo), $\sigma$ (combo)                 | 8.874, 5.000              | Derived         | Ratio fit                |
| r_LR $\mu$ (pembro), $\sigma$ (pembro)               | 5.913, 2.674              | Derived         | Ratio fit                |
| <b>Long-term extrapolation</b>                       |                           |                 |                          |
| 10-year OS, combo                                    | 84.1%                     | Modeled         | Log-normal extrapolation |
| 10-year OS, pembro                                   | 51.6%                     | Modeled         | Log-normal extrapolation |
| 20-year OS, combo                                    | 60.1%                     | Modeled         | GP hazard floor          |
| (GP-constrained)                                     |                           |                 |                          |
| 20-year OS, pembro                                   | 28.5%                     | Modeled         | GP hazard floor          |
| (GP-constrained)                                     |                           |                 |                          |
| 30-year OS, combo                                    | 22.2%                     | Modeled         | GP hazard floor          |
| (GP-constrained)                                     |                           |                 |                          |
| 30-year OS, pembro                                   | 10.5%                     | Modeled         | GP hazard floor          |
| (GP-constrained)                                     |                           |                 |                          |
| 40-year OS, combo                                    | 2.0%                      | Modeled         | GP hazard floor          |
| (GP-constrained)                                     |                           |                 |                          |

### 3.1 Observed vs. Fitted Survival Curves



Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm.

## 4 Survival Model Selection

### 4.1 Component-Level Parametric Fit

Table S4: Component-Level Parametric Fit Across Five Distributions

| Component                          | Distribution | AIC    | RMSE   | Rank |
|------------------------------------|--------------|--------|--------|------|
| <i>OS – Combo arm</i>              |              |        |        |      |
| Exponential                        | 1 param      | -45.49 | 0.0087 | 4    |
| Weibull                            | 2 param      | -49.84 | 0.0046 | 2    |
| Log-normal                         | 2 param      | -50.08 | 0.0045 | 1    |
| Log-logistic                       | 2 param      | -49.89 | 0.0046 | 3    |
| Generalized gamma                  | 3 param      | -47.97 | 0.0045 | 5    |
| <i>OS – Pembro arm</i>             |              |        |        |      |
| Exponential                        | 1 param      | -31.27 | 0.0359 | 4    |
| Weibull                            | 2 param      | -32.00 | 0.0273 | 1    |
| Log-normal                         | 2 param      | -31.16 | 0.0297 | 3    |
| Log-logistic                       | 2 param      | -31.69 | 0.0282 | 2    |
| Generalized gamma                  | 3 param      | -30.33 | 0.0264 | 5    |
| <i>r<sub>DM</sub> – Combo arm</i>  |              |        |        |      |
| Exponential                        | 1 param      | -36.79 | 0.0078 | 3    |
| Weibull                            | 2 param      | -36.53 | 0.0063 | 4    |
| Log-normal                         | 2 param      | -36.93 | 0.0060 | 1    |
| Log-logistic                       | 2 param      | -36.61 | 0.0062 | 2    |
| Generalized gamma                  | 3 param      | -34.74 | 0.0061 | 5    |
| <i>r<sub>DM</sub> – Pembro arm</i> |              |        |        |      |
| Exponential                        | 1 param      | -19.41 | 0.0688 | 5    |
| Weibull                            | 2 param      | -28.43 | 0.0174 | 3    |
| Log-normal                         | 2 param      | -27.23 | 0.0202 | 4    |
| Log-logistic                       | 2 param      | -30.30 | 0.0137 | 2    |
| Generalized gamma                  | 3 param      | -32.45 | 0.0082 | 1    |
| <i>r<sub>LR</sub> – Combo arm</i>  |              |        |        |      |
| Exponential                        | 1 param      | -30.38 | 0.0392 | 5    |
| Weibull                            | 2 param      | -38.97 | 0.0136 | 3    |
| Log-normal                         | 2 param      | -39.49 | 0.0129 | 2    |
| Log-logistic                       | 2 param      | -39.70 | 0.0126 | 1    |
| Generalized gamma                  | 3 param      | -38.43 | 0.0118 | 4    |
| <i>r<sub>LR</sub> – Pembro arm</i> |              |        |        |      |
| Exponential                        | 1 param      | -31.82 | 0.0340 | 5    |
| Weibull                            | 2 param      | -33.60 | 0.0233 | 1    |
| Log-normal                         | 2 param      | -33.37 | 0.0238 | 2    |
| Log-logistic                       | 2 param      | -33.51 | 0.0235 | 3    |
| Generalized gamma                  | 3 param      | -31.69 | 0.0231 | 4    |

### 4.2 Joint Model Selection

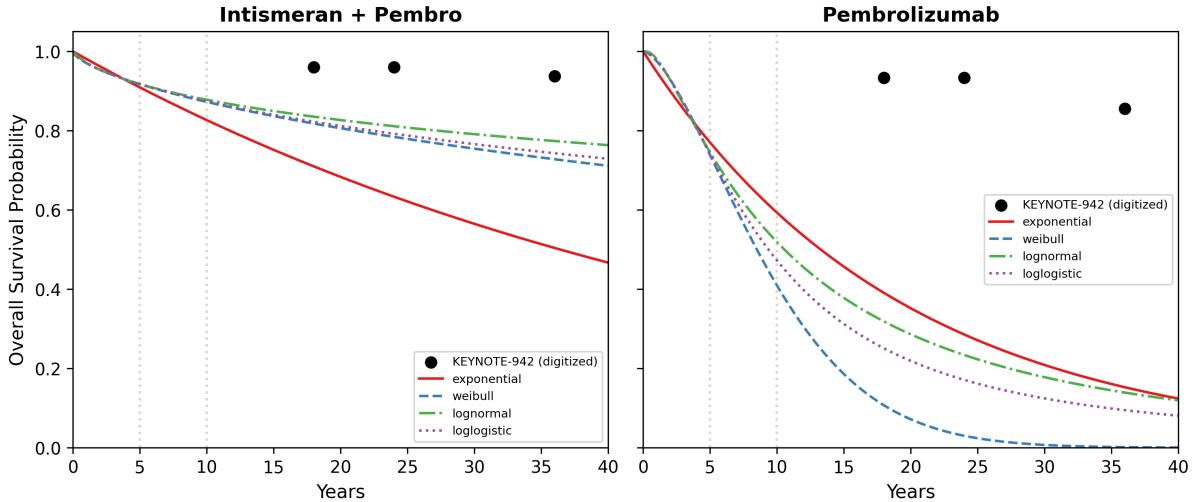
A total of 15,625 distribution combinations (5 distributions x 6 curves) were evaluated for structural validity at 40 years. All 15,625 satisfied the  $RFS \leq DMFS \leq OS$  ordering constraint (0 crossing violations). The full results are available online ([output/survival\\_combos\\_15625.csv](#)).

Table S5: Top 10 Joint Model Combinations by Total AIC (5 distributions, 15,625 combinations)

| Rank | OS       | r <sub>DM</sub> | r <sub>LR</sub> | Total AIC | OS40c | OS40p |
|------|----------|-----------------|-----------------|-----------|-------|-------|
|      | C/P      | C/P             | C/P             |           |       |       |
| 1    | LN / W   | LN / GG         | LL / W          | -224.76   | 76.3% | 0.0%  |
| 2    | LN / W   | LN / GG         | LL / LL         | -224.67   | 76.3% | 0.0%  |
| 3    | LN / W   | Exp / GG        | LL / W          | -224.62   | 76.3% | 0.0%  |
| 4    | LLog / W | LN / GG         | LL / W          | -224.57   | 72.9% | 0.0%  |
| 5    | LN / W   | LN / GG         | LN / W          | -224.55   | 76.3% | 0.0%  |
| 6    | LN / W   | LN / GG         | LL / LN         | -224.53   | 76.3% | 0.0%  |
| 7    | LN / W   | Exp / GG        | LL / LL         | -224.53   | 76.3% | 0.0%  |
| 8    | W / W    | LN / GG         | LL / W          | -224.52   | 71.1% | 0.0%  |
| 9    | LLog / W | LN / GG         | LL / LL         | -224.48   | 72.9% | 0.0%  |
| 10   | LN / W   | LN / GG         | LN / LL         | -224.46   | 76.3% | 0.0%  |

*Distributions:* LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential, GG = generalized gamma. C/P = Combo/Pembro. All top-ranked combinations use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3% under unconstrained extrapolation, substantially higher than GP survival at age 101 (2.0%). This motivated the GP mortality hazard floor (Section 5.1).

### 4.3 Lifetime OS Extrapolation by Distribution



Vertical dotted lines at 5 and 10 years mark the boundary between within-trial and extrapolated periods.

Figure S3: Lifetime OS extrapolation by parametric distribution. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; beyond 10 years is pure extrapolation. The four distributions produce wildly divergent 40-year survival estimates (0% to 76.3%).

## 5 General-Population Mortality and Structural Validation

### 5.1 GP Mortality Implementation

The GP mortality hazard floor is applied to modeled OS from month 60 onward.

**Life table source:** US 2019 period life tables (Arias et al., NVSS). **Baseline age:** 61 years (KEYNOTE-942 median). **Sex weighting:** 64% male, 36% female. **Interpolation:** Annual life table survival linearly interpolated to monthly resolution; monthly hazards derived as  $h_{GP}(t) = -\ln[S_{GP}(t+1)/S_{GP}(t)]$ . **Constraint formula (mortality hazard floor):** from month 60 onward, the model mortality hazard

is floored at the age- and sex-matched general-population hazard:

$$h_{\text{model}}^{\text{con}}(t) = \max[h_{\text{param}}(t), h_{\text{GP}}(t)] \quad t \geq 60 \text{ months}, \quad (1)$$

where  $h_{\text{param}}(t) = -\ln[S_{\text{param}}(t+1)/S_{\text{param}}(t)]$  is the monthly hazard implied by the fitted parametric survival curve. The constrained survival is reconstructed as  $S_{\text{model}}^{\text{con}}(t) = \exp[-\sum_0^t h^{\text{con}}(u)]$ . This is a **hazard floor**, not a survival cap: a cohort that has already experienced elevated (cancer-related) mortality in the first 5 years remains below general-population survival at every subsequent time point, whereas a survival cap ( $S = \min[S_{\text{param}}, S_{\text{GP}}]$ ) would incorrectly force the two curves to coincide whenever the parametric tail exceeded the general-population level. Both arms are subject to the same background mortality floor.

## 5.2 Long-Term Survival Diagnostics

Table S6: Long-Term Survival Predictions vs. General Population. GP survival is conditional on being alive at baseline age 61 and is calculated as  $S_{\text{GP}}(t) = l_{61+t}/l_{61}$  using the sex-weighted 2019 US life tables (64% male, 36% female); the terminal 100+ life-table value is used for the 40-year row.

| Years from baseline | Attained age | Combo OS | Pembro OS | GP survival |
|---------------------|--------------|----------|-----------|-------------|
| 5                   | 66           | 91.6%    | 74.1%     | 94.1%       |
| 10                  | 71           | 84.1%    | 51.6%     | 86.4%       |
| 15                  | 76           | 74.0%    | 37.6%     | 76.1%       |
| 20                  | 81           | 60.1%    | 28.5%     | 61.9%       |
| 30                  | 91           | 22.2%    | 10.5%     | 23.1%       |
| 40                  | 101          | 2.0%     | 0.9%      | 2.0%        |

## 5.3 OS versus General-Population Survival



GP-constrained curves (solid) use a mortality hazard floor from month 60. The constrained combo OS stays below the GP curve at all time points, reflecting the excess mortality experienced in the first 5 years. Unconstrained log-normal (dotted) produces implausibly high 40-year survival (76.4%).

Figure S4: OS versus general population. Solid lines: GP-constrained model (hazard floor from 60 months). Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US GP survival from age 61. The constrained combo curve remains below the GP curve at all time points, reflecting the hazard-floor property: a cohort with prior excess mortality does not "catch up" to general-population survival.

## 5.4 Structural Quality Assurance

Table S7: Structural Quality Assurance (480 Monthly Cycles)

| Check                              | Result       |
|------------------------------------|--------------|
| Total cycles                       | 480          |
| RFS > DMFS violations              | 0            |
| DMFS > OS violations               | 0            |
| Negative RF state probability      | 0            |
| Negative LR state probability      | 0            |
| Negative DM state probability      | 0            |
| Maximum state-sum error            | $< 10^{-12}$ |
| Clamp events (forced non-negative) | 0            |

## 6 External Validation

### 6.1 External Validation Landmarks

Table S8: External Validation of Pembrolizumab Arm Predictions

| Outcome | Year | Model (pembro) | KEYNOTE-054 | CheckMate 238 |
|---------|------|----------------|-------------|---------------|
| RFS     | 5    | 40.5%          | —           | 51% [? ]      |
| RFS     | 7    | 31.9%          | 50% [? ]    | 46% [? ]      |
| DMFS    | 5    | 53.8%          | —           | 59% [? ]      |
| DMFS    | 7    | 44.9%          | 54% [? ]    | 55% [? ]      |
| OS      | 5    | 74.5%          | NR          | 76% [? ]      |
| OS      | 7    | 64.2%          | NR          | 71% [? ]      |

NR: Not reported. CheckMate 238 evaluated nivolumab (not pembrolizumab). The model predicts lower RFS/DMFS versus external trials, reflecting the higher-risk KEYNOTE-942 population (stage IIIB–IV including resected stage IV melanoma vs IIIA–C in KEYNOTE-054 and IIIB–IV in CheckMate 238). Only published landmark values are included; no extrapolated estimates are used.[? ? ]

### 6.2 Longitudinal External Validation



Figure S5: External validation of pembrolizumab arm predictions. Gray line: model. Circles: KEYNOTE-942 observed data. Squares: KEYNOTE-054 (pembrolizumab, 7-year). Triangles: CheckMate 238 (nivolumab, 5- and 7-year). Only published landmark values are shown.

## 7 Model Inputs

### 7.1 Complete Parameter Table

Table S9: Full Model Parameter Inventory

| Parameter                         | Base                | DSA Low   | DSA High  | PSA Distribution          | Source            |
|-----------------------------------|---------------------|-----------|-----------|---------------------------|-------------------|
| <i>Clinical Parameters</i>        |                     |           |           |                           |                   |
| OS $\mu$ – combo                  | 8.396               | 5.038     | 11.754    | Normal (SE from Jacobian) | Fitted            |
| OS $\sigma$ – combo               | 3.096               | 2.896     | 3.296     | Normal (SE from Jacobian) | Fitted            |
| OS $\mu$ – pembro                 | 4.841               | 3.873     | 5.809     | Normal (SE from Jacobian) | Fitted            |
| OS $\sigma$ – pembro              | 1.131               | 0.931     | 1.331     | Normal (SE from Jacobian) | Fitted            |
| r <sub>DM</sub> $\mu$ – combo     | 7.113               | 6.913     | 7.313     | Normal (SE from Jacobian) | Fitted            |
| r <sub>DM</sub> $\sigma$ – combo  | 2.480               | 2.280     | 2.680     | Normal (SE from Jacobian) | Fitted            |
| r <sub>DM</sub> $\mu$ – pembro    | 7.042               | 6.842     | 7.242     | Normal (SE from Jacobian) | Fitted            |
| r <sub>DM</sub> $\sigma$ – pembro | 5.000               | 4.800     | 5.200     | Normal (SE from Jacobian) | Fitted            |
| r <sub>LR</sub> $\mu$ – combo     | 8.874               | 8.674     | 9.074     | Normal (SE from Jacobian) | Fitted            |
| r <sub>LR</sub> $\sigma$ – combo  | 5.000               | 4.800     | 5.200     | Normal (SE from Jacobian) | Fitted            |
| r <sub>LR</sub> $\mu$ – pembro    | 5.913               | 5.713     | 6.113     | Normal (SE from Jacobian) | Fitted            |
| r <sub>LR</sub> $\sigma$ – pembro | 2.674               | 2.474     | 2.874     | Normal (SE from Jacobian) | Fitted            |
| Treatment duration (pembro)       | 18 cycles (2 yr)    | 9         | 24        | Fixed                     | [? ]              |
| Treatment duration (intismeran)   | 9 doses (1 yr)      | 6         | 12        | Fixed                     | [? ]              |
| Baseline age                      | 61 years            | 55        | 67        | Fixed                     | [? ]              |
| Male proportion                   | 64%                 | –         | –         | Fixed                     | [? ]              |
| <i>Costs (2026 USD)</i>           |                     |           |           |                           |                   |
| Intismeran (one-time)             | \$200,000           | \$160,000 | \$240,000 | Gamma (CV=0.2)            | Assumed           |
| Pembrolizumab (annual)            | \$220,896           | \$176,717 | \$265,075 | Gamma (CV=0.2)            | WAC (CMS)         |
| Tumor sequencing                  | \$1,000             | \$500     | \$2,000   | Gamma (CV=0.2)            | CMS               |
| Administration (per cycle)        | \$500               | \$250     | \$750     | Gamma (CV=0.2)            | CMS               |
| AE management (one-time)          | \$15,000            | \$7,500   | \$22,500  | Gamma (CV=0.2)            | [? ]              |
| LR management (monthly)           | \$3,000             | \$2,400   | \$3,600   | Gamma (CV=0.2)            | [? ]              |
| DM management (monthly)           | \$12,000            | \$9,600   | \$14,400  | Gamma (CV=0.2)            | [? ]              |
| <i>Utilities</i>                  |                     |           |           |                           |                   |
| RF state                          | 0.83                | 0.80      | 0.86      | Beta (SE=0.03)            | [? ]              |
| LR state                          | 0.64                | 0.61      | 0.67      | Beta (SE=0.03)            | [? ]              |
| DM state                          | 0.55                | 0.52      | 0.58      | Beta (SE=0.03)            | [? ]              |
| AE disutility (one-time)          | 0.05                | 0.00      | 0.10      | Beta (SE=0.02)            | Assumed           |
| <i>Economic Parameters</i>        |                     |           |           |                           |                   |
| Discount rate (costs, effects)    | 3%                  | 0%        | 5%        | Fixed (DSA only)          | [? ]              |
| Time horizon                      | 40 years            | 10        | 20        | Fixed (scenario)          | Assumed           |
| Willingness-to-pay threshold      | \$100K–\$150K/QALY  | \$50K     | \$200K    | Fixed (scenario)          | [? ]              |
| Cost base year                    | 2026 USD            | –         | –         | –                         | Assumed           |
| GP mortality source               | US 2019 Life Tables | –         | –         | –                         | NVSS (Arias 2022) |

## 8 Additional Results

### 8.1 Lifetime Cost Decomposition

Table S10: Lifetime Cost Decomposition by Component (GP-constrained, 40-year horizon)

| Component      | Combo            | Pembro           | Incremental      |
|----------------|------------------|------------------|------------------|
| Intismeran     | \$200,000        | \$0              | \$200,000        |
| Pembrolizumab  | \$217,888        | \$217,888        | \$0              |
| Sequencing     | \$1,000          | \$0              | \$1,000          |
| Administration | \$493            | \$493            | \$0              |
| Adverse events | \$15,000         | \$0              | \$15,000         |
| LR management  | \$85,455         | \$69,341         | \$16,114         |
| DM management  | \$354,189        | \$424,388        | \$-70,199        |
| <b>Total</b>   | <b>\$874,026</b> | <b>\$712,110</b> | <b>\$161,916</b> |

### 8.2 LY and QALY Decomposition

Table S11: Life-Year and QALY Decomposition by Health State

| Outcome                    | Combo        | Pembro      | Incremental |
|----------------------------|--------------|-------------|-------------|
| Life-years (undiscounted)  | 21.58        | 13.81       | 7.77        |
| Life-years (discounted 3%) | 15.06        | 10.28       | 4.78        |
| RF state QALYs             | 8.49         | 4.48        | 4.01        |
| LR state QALYs             | 1.52         | 1.23        | 0.29        |
| DM state QALYs             | 1.35         | 1.62        | -0.27       |
| <b>Total QALYs</b>         | <b>11.35</b> | <b>7.33</b> | <b>4.02</b> |

Health benefit is driven by RF state QALYs (4.01 gained), with a modest additional contribution from LR (+0.29) and a small reduction in DM state QALYs (−0.27) under the corrected mortality-constrained extrapolation.

## 9 Uncertainty and Structural Scenario Analyses

### 9.1 Deterministic Sensitivity Analysis

Table S12: One-Way Deterministic Sensitivity Analysis

| Parameter                 | Base      | Low       | High      | ICER range      | Change in NMB |
|---------------------------|-----------|-----------|-----------|-----------------|---------------|
| Utility RF                | 0.83      | 0.80      | 0.86      | 41,777–38,875   | \$43,396      |
| Utility LR                | 0.64      | 0.61      | 0.67      | 40,409–40,140   | \$4,029       |
| Utility DM                | 0.55      | 0.52      | 0.58      | 40,128–40,421   | \$4,387       |
| Pembrolizumab annual cost | \$220,896 | \$176,717 | \$265,075 | 40,274–40,274   | \$0           |
| Intismeran cost           | \$200,000 | \$160,000 | \$240,000 | 30,324–50,223   | \$80,000      |
| LR monthly cost           | \$3,000   | \$2,400   | \$3,600   | 39,472–41,075   | \$6,446       |
| DM monthly cost           | \$12,000  | \$9,600   | \$14,400  | 43,766–36,782   | \$28,079      |
| Discount rate             | 3%        | 0%        | 5%        | 33,120–48,284   | \$434,030     |
| OS $\mu$ combo            | 8.396     | -40%      | +40%      | 19,875–40,684   | \$485,457     |
| OS $\mu$ pembro           | 4.841     | -20%      | +20%      | 57,434–Dominant | \$441,968     |

The maximum finite ICER was \$57,434/QALY across the tested ranges; the combination became dominant when the pembrolizumab OS parameter was increased by 20%.

## **9.2 Probabilistic Sensitivity Analysis**

### **9.2.1 PSA Implementation Details**

PSA with 6,000 Monte Carlo iterations (seed = 42). Survival parameters ( $\mu$ ,  $\sigma$  for 6 curves) drawn jointly from bivariate normal via Jacobian covariance ( $r > 0.95$ ). Costs: gamma (CV=0.2). Utilities: beta (SE=0.03). Structural constraints ( $RFS \leq DMFS \leq OS$ , monotonicity, survival within  $[0,1]$ ) verified every cycle; all 6,000 draws were structurally valid (0 rejections). No iterations were excluded on the basis of the ICER value.

Table S13: PSA Parameter Distributions and Sampled Values

| Parameter                                 | Distribution     | Fitted mean        | SE            | Correlation | Sampling            |
|-------------------------------------------|------------------|--------------------|---------------|-------------|---------------------|
| OS $\mu$ / $\sigma$ – combo               | Bivariate normal | 8.396 / 3.096      | 0.793 / 0.522 | 0.996       | Joint               |
| OS $\mu$ / $\sigma$ – pembro              | Bivariate normal | 4.841 / 1.131      | 0.291 / 0.310 | 0.955       | Joint               |
| r <sub>DM</sub> $\mu$ / $\sigma$ – combo  | Bivariate normal | 7.113 / 2.480      | 0.814 / 0.563 | 0.995       | Joint               |
| r <sub>DM</sub> $\mu$ / $\sigma$ – pembro | Bivariate normal | 7.042 / 5.000      | 2.278 / 3.130 | 0.995       | Joint               |
| r <sub>LR</sub> $\mu$ / $\sigma$ – combo  | Bivariate normal | 8.874 / 5.000      | 2.039 / 1.928 | 0.997       | Joint               |
| r <sub>LR</sub> $\mu$ / $\sigma$ – pembro | Bivariate normal | 5.913 / 2.674      | 0.757 / 0.879 | 0.984       | Joint               |
| Intismeran cost                           | Fixed            | \$200,000          | —             | —           | Fixed (not sampled) |
| Pembrolizumab annual cost                 | Gamma (CV=0.2)   | \$220,896          | —             | —           | Independent         |
| LR monthly cost                           | Gamma (CV=0.2)   | \$3,000            | —             | —           | Independent         |
| DM monthly cost                           | Gamma (CV=0.2)   | \$12,000           | —             | —           | Independent         |
| Utility RF / LR / DM                      | Beta (SE=0.03)   | 0.83 / 0.64 / 0.55 | —             | —           | Independent         |

### 9.2.2 PSA Results

- Mean  $\Delta QALY$ : 4.32 (95% CI 2.53–7.04).
- Mean  $\Delta Cost$ : \$127,711 (95% CI –\$668,394 to \$520,672).
- Mean NMB: \$304,291 at \$100K and \$520,293 at \$150K.
- Probability cost-effective: 63.8% at \$50K, 95.3% at \$100K, and 98.7% at \$150K.
- No iterations were excluded based on the ICER; 16.7% of draws were cost-saving, and all 6,000 draws had  $\Delta QALY > 0$ .

### 9.2.3 PSA Convergence



Figure S6: PSA convergence across iteration counts (6,000 structurally valid iterations, 0 structural rejections, seed = 42). Cumulative means of  $\Delta QALY$ ,  $\Delta Cost$ , NMB, and P(CE). Vertical dotted lines at 500, 1,000, 2,000, and 4,000 iterations. Estimates stable from ~500 iterations.

### 9.2.4 Cost-Effectiveness Plane

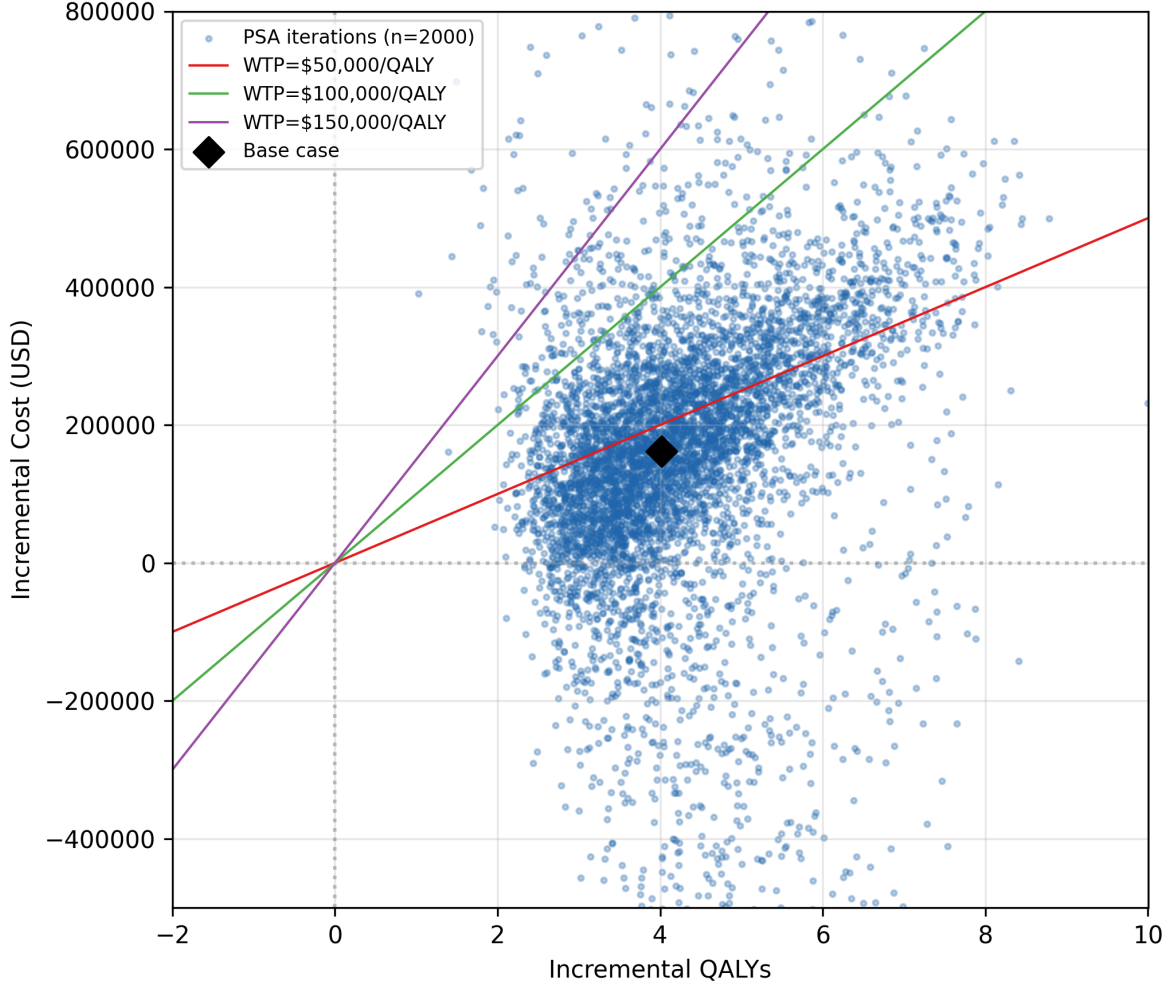


Figure S7: CE plane showing 6,000 PSA iterations (blue), WTP lines (\$50K, \$100K, \$150K per QALY), and the deterministic base case (black diamond). 16.7% of iterations fall in the southeast quadrant (cost-saving); all 6,000 iterations had positive  $\Delta$ QALY. No iterations were excluded based on the ICER value.

### 9.3 Structural Scenario Analyses

**Scenario implementations.** Three structural scenarios alter the base-case treatment-effect assumptions. All scenarios preserve the observed 5-year KEYNOTE-942 data unchanged and modify only the extrapolated trajectory beyond 60 months.

*OS-survival convergence (5–20 y, Waning V1).* On the **survival probability scale**, the incremental OS benefit (combo OS minus pembrolizumab OS) is linearly attenuated from its full base-case value at 60 months to zero at 240 months (20 years). At each monthly cycle  $t$ :  $S_{\text{combo}}^{\text{waned}}(t) = S_{\text{pembro}}(t) + [S_{\text{combo}}^{\text{base}}(t) - S_{\text{pembro}}(t)] \times \max(0, 1 - \frac{t-60}{180})$ . The recurrence ( $r_{\text{DM}}$ ) and local-recurrence ( $r_{\text{LR}}$ ) ratio functions are **not** modified, so the scenario isolates the effect of attenuating the OS benefit while preserving the observed recurrence advantage. The general-population hazard floor continues to apply independently to both arms.

*Hazard convergence (5–10 y, Waning V2).* On the **cause-specific mortality hazard scale**, the combo arm’s monthly all-cause hazard is linearly blended toward the pembrolizumab arm’s hazard from month 60 to month 120 (10 years). At each cycle:  $h_{\text{combo}}^{\text{waned}}(t) = (1 - w(t))h_{\text{combo}}(t) + w(t)h_{\text{pembro}}(t)$ , where  $w(t) = \max(0, \min(\frac{t-60}{60}, 1))$ . The waned hazard is integrated forward to reconstruct the survival curve:  $S_{\text{combo}}^{\text{waned}}(t+1) = S_{\text{combo}}^{\text{waned}}(t) \times \exp(-h_{\text{combo}}^{\text{waned}}(t))$ , with  $S_{\text{combo}}^{\text{waned}}(t) = S_{\text{combo}}^{\text{base}}(t)$  for  $t \leq 60$  months. In this

scenario,  $r_{\text{DM}}$  and  $r_{\text{LR}}$  are also linearly converged to pembrolizumab-arm values on the probability scale ( $\text{HR} \rightarrow 1$  by 10 years), so that all three survival components reflect complete treatment-effect erosion. The GP hazard floor continues to apply.

*No direct OS benefit.* The combination arm's OS curve is set equal to the pembrolizumab arm's OS curve at all time points ( $S_{\text{combo}}(t) = S_{\text{pembro}}(t) \forall t$ ). *Therecurrence*( $r_{\text{DM}}$ ) and local-recurrence ( $r_{\text{LR}}$ ) ratio functions retain their base-case values, so any incremental QALY gain arises solely from the improved disease-state trajectory (longer recurrence-free and distant-metastasis-free intervals) rather than from extended overall survival.  $\text{RFS} \leq \text{DMFS} \leq \text{OS}$  ordering is maintained trivially since both arms share the same OS curve.

Table S14: Structural Scenario Analysis Results

| Scenario                            | $\Delta LY$ | $\Delta QALY$ | $\Delta Cost$ | ICER      | NMB \$100K | NMB \$150K |
|-------------------------------------|-------------|---------------|---------------|-----------|------------|------------|
| Base case                           | 4.78        | 4.02          | \$161,916     | \$40,274  | \$240,123  | \$441,143  |
| Weibull OS                          | 5.60        | 5.60          | \$307,578     | \$54,925  | \$252,422  | \$532,422  |
| Unconstrained (no GP)               | 9.24        | 7.22          | \$373,488     | \$51,753  | \$348,512  | \$709,512  |
| GP floor 48 mo                      | 4.78        | 4.00          | \$160,954     | \$40,236  | \$239,046  | \$439,046  |
| GP floor 60 mo (base)               | 4.78        | 4.02          | \$161,916     | \$40,274  | \$240,123  | \$441,143  |
| GP floor 72 mo                      | 4.78        | 4.05          | \$163,611     | \$40,351  | \$241,389  | \$443,889  |
| GP floor 96 mo                      | 4.91        | 4.15          | \$168,745     | \$40,623  | \$246,255  | \$453,755  |
| 10-year horizon                     | 1.30        | 1.30          | \$96,400      | \$74,305  | \$33,600   | \$98,600   |
| 20-year horizon                     | 3.54        | 3.08          | \$123,974     | \$40,293  | \$183,705  | \$337,545  |
| 30-year horizon                     | 3.89        | 3.89          | \$155,242     | \$39,914  | \$233,758  | \$428,258  |
| 40-year horizon (base)              | 4.78        | 4.02          | \$161,916     | \$40,274  | \$240,123  | \$441,143  |
| Intismeran \$100K                   | 4.78        | 4.02          | \$61,916      | \$15,400  | \$340,123  | \$541,143  |
| Intismeran \$200K (base)            | 4.78        | 4.02          | \$161,916     | \$40,274  | \$240,123  | \$441,143  |
| Intismeran \$300K                   | 4.78        | 4.02          | \$261,916     | \$65,147  | \$140,123  | \$341,143  |
| Intismeran \$500K                   | 4.78        | 4.02          | \$461,916     | \$114,893 | -\$59,877  | \$141,143  |
| Waning V1 (OS-survival convergence) | 1.91        | 1.93          | \$41,571      | \$21,514  | \$151,429  | \$248,929  |
| Waning V2 (hazard convergence)      | 2.72        | 2.17          | \$227,542     | \$104,852 | -\$10,530  | \$97,975   |
| No direct OS benefit                | 0.00        | 0.50          | -\$14,330     | Dominant  | \$64,330   | \$89,330   |

The base-case conclusion is robust across structural assumptions. The ICER alone is insufficient: unconstrained model has moderate ICER (\$51,753) but implausible \$373,488 cost.

**Time horizon sensitivity.** The ICER remained below conventional US willingness-to-pay thresholds across 20-, 30-, and 40-year horizons, although the point estimate increased at the 40-year horizon: \$40,293 at 20 years, \$39,914 at 30 years, and \$40,274 at the 40-year base-case horizon (Table S14). The 30-to-40-year increase of \$360/QALY (0.9%) reflects the limited contribution of late discounted QALY gains beyond 30 years; the 10-year horizon produced a higher ICER of \$74,305/QALY, reflecting the incomplete capture of long-term QALY gains within a shorter window. Because the general-population mortality hazard floor was applied from 60 months onward and both costs and QALYs were discounted at 3% annually, survival beyond the 20-year window contributed materially to discounted incremental costs and QALYs.

**General population hazard floor start-time sensitivity.** The GP mortality floor was applied from 60 months in the base case. Varying the onset from 48 to 96 months produced no material change in the ICER (the ICERs ranged from \$40,236 to \$40,623/QALY across 48–96 months). This reflects the fact that the fitted hazard for the combination arm closely tracks the general-population mortality hazard in the post-trial period (both at approximately 0.0008–0.0009 monthly), leaving the floor with minimal additional constraining effect regardless of onset timing. The floor’s binding is thus a near-numerical coincidence between the parametric extrapolation and US life-table mortality, not a model artifact.

## 10 Deterministic Threshold Acquisition Price

### 10.1 Price Threshold Analysis

Table S15: Price Threshold Analysis

| Intismeran Price | ICER             | NMB @ \$100K      | NMB @ \$150K     |
|------------------|------------------|-------------------|------------------|
| \$0              | Dominant         | \$440,123         | \$641,143        |
| \$50,000         | \$2,964          | \$390,123         | \$591,143        |
| \$100,000        | \$15,400         | \$340,123         | \$541,143        |
| \$150,000        | \$27,837         | \$290,123         | \$491,143        |
| \$200,000 (base) | \$40,274         | \$240,123         | \$441,143        |
| \$250,000        | \$52,710         | \$190,123         | \$391,143        |
| \$300,000        | \$65,147         | \$140,123         | \$341,143        |
| \$350,000        | \$77,583         | \$90,123          | \$291,143        |
| \$400,000        | \$90,020         | \$40,123          | \$241,143        |
| \$450,000        | \$102,457        | -\$9,877          | \$191,143        |
| \$500,000        | \$114,893        | -\$59,877         | \$141,143        |
| <b>\$440,063</b> | <b>\$100,000</b> | <b>\$0</b>        | <b>\$200,000</b> |
| \$550,000        | \$127,330        | -\$109,877        | \$91,143         |
| \$600,000        | \$139,766        | -\$159,877        | \$41,143         |
| \$650,000        | \$152,203        | -\$209,877        | -\$8,857         |
| \$700,000        | \$164,640        | -\$259,877        | -\$58,857        |
| \$750,000        | \$177,076        | -\$309,877        | -\$108,857       |
| \$800,000        | \$189,513        | -\$359,877        | -\$158,857       |
| <b>\$640,869</b> | <b>\$150,000</b> | <b>-\$200,746</b> | <b>\$0</b>       |

Deterministic threshold prices (from Table S15): approximately \$440,063 at \$100,000/QALY (2.2× base) and approximately \$640,869 at \$150,000/QALY (3.2× base). These are model-based deterministic estimates; PSA uncertainty propagation and formal budget-impact analysis would be needed to inform payer negotiations.

## 10.2 ICER versus Acquisition Price

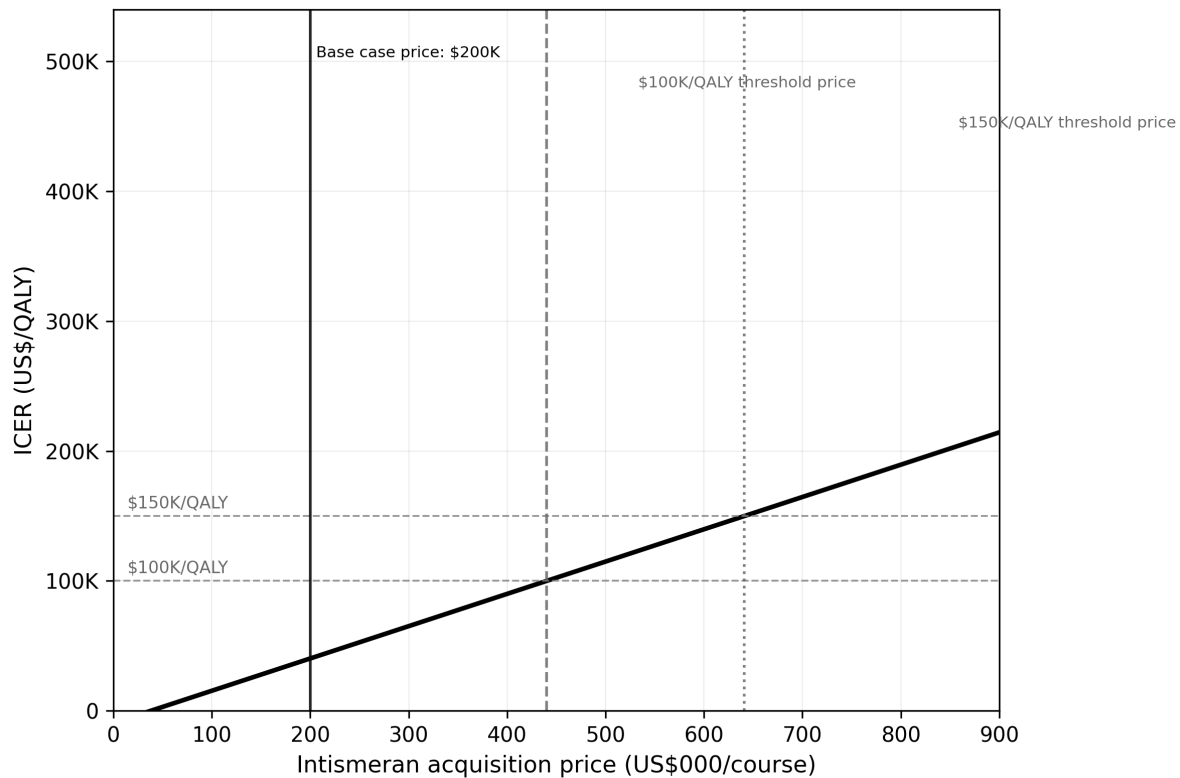


Figure S8: ICER as a function of intismeran acquisition price. Red/green dashed lines: \$100K/\$150K thresholds. Vertical dotted lines: threshold prices.

## 11 Reproducibility and Reporting

### 11.1 Technical Reproducibility

All analyses were performed using Python 3.12.7 (Apple Clang 15.0.0) on macOS 26.6.2 (Apple Silicon). The full code repository is available at <https://github.com/yichao2022/cea-intismeran> (tag v1.0-submission).

**Key dependencies:** NumPy 1.26.4, SciPy 1.14.1, Matplotlib 3.9.3, Pandas 2.2.3, LaTeX (TeX Live 2025, pdfLaTeX + BibTeX + natbib).

**Model details:**

- Cycle length: monthly (1/12 year); time horizon: 40 years (480 cycles)
- Half-cycle correction: not applied (monthly cycles are sufficiently fine-grained)
- Discounting: continuous-time exponential:  $1/(1+r)^t$  where  $r = 0.03$
- Random seed: 42 (PSA reproducibility)
- GP mortality source: US 2019 period life tables (NVSS, Arias et al. 2022)

**Code structure:**

- `model.py` — ModelParams, CEAModel (4-state PSM)
- `rerun_primary.py` — CEAModelV2 (hazard-based GP floor, waning, no-OS)
- `psa_full.py` — PSA with joint survival sampling
- `survival_model_selection.py` — 15,625-combination model selection
- `final_run.py` / `threshold_analysis.py` — DSA, scenarios, pricing

- `figures.py` / `gen_fig*.py` — Figure generation scripts
- `output/survival_combos_15625.csv` — Full model selection results

**Data extraction:** Survival probabilities extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7. The digitized values are in Table S2.

**Reproducibility:** Running `python3 rerun_primary.py` reproduces base-case results. `python3 psa.full.py` reproduces PSA. GitHub tag `v1.0-submission` freezes the submission code.

## 11.2 CHEERS 2022 Checklist

Table S16: CHEERS 2022 Checklist

| No. | Item                          | Reported     | Location                                                   |
|-----|-------------------------------|--------------|------------------------------------------------------------|
| 1   | Title                         | Yes          | Title page                                                 |
| 2   | Abstract                      | Yes          | Abstract                                                   |
| 3   | Background                    | Yes          | Introduction                                               |
| 4   | Health economic analysis plan | No           | This is a retrospective analysis, not a pre-specified HEAP |
| 5   | Model                         | Yes          | Methods, Appendix A                                        |
| 6   | Population                    | Yes          | Methods                                                    |
| 7   | Setting                       | Yes          | Methods                                                    |
| 8   | Comparators                   | Yes          | Methods                                                    |
| 9   | Perspective                   | Yes          | Methods                                                    |
| 10  | Time horizon                  | Yes          | Methods                                                    |
| 11  | Discount rate                 | Yes          | Methods                                                    |
| 12  | Outcomes                      | Yes          | Methods                                                    |
| 13  | Health states                 | Yes          | Appendix A                                                 |
| 14  | Measurement of outcomes       | Yes          | Methods                                                    |
| 15  | Resource use                  | Yes          | Methods                                                    |
| 16  | Currency                      | Yes          | Methods                                                    |
| 17  | Model choice                  | Yes          | Methods                                                    |
| 18  | Assumptions                   | Yes          | Methods (§2.2–2.7)                                         |
| 19  | Analytic methods              | Yes          | Methods (§2.7)                                             |
| 20  | Study parameters              | Yes          | Methods (§2.4, 2.5), Appendix D                            |
| 21  | Incremental results           | Yes          | Results (§3.1–3.2)                                         |
| 22  | Uncertainty                   | Yes          | Results (§3.4–3.6), Supplement                             |
| 23  | Heterogeneity                 | Not assessed | Limitations (§4.7, subgroup heterogeneity acknowledged)    |
| 24  | Discussion                    | Yes          | Discussion                                                 |
| 25  | Limitations                   | Yes          | Discussion                                                 |
| 26  | Funding                       | Yes          | Declarations                                               |
| 27  | Conflicts                     | Yes          | Declarations                                               |
| 28  | Reproducibility               | Yes          | Appendix                                                   |